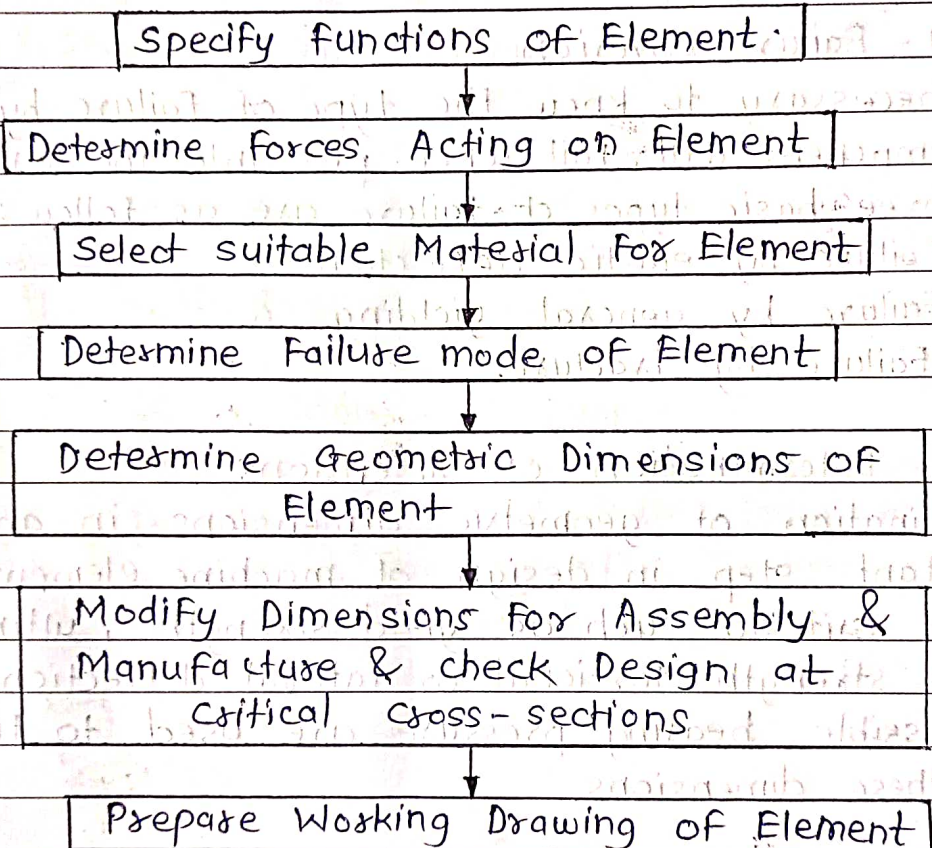


Assignment No. 01 - Mechanical Engineering Design Process

Q.1 state & explain various steps involved in design of machine element.

→ The basic procedure of the design of machine elements is illustrated in below figure.



Step 1: Specification of Function

- The design of machine element begins with the specification of the functions of the element.
- The functions of some machine elements are as follows:
 - i) **Beasing**: to support the rotating shaft & confine its motion.
 - ii) **Power Screw**: to produce uniform & slow motion & to transmit the force.

Step 2 - Determination of Forces

- In many cases, a free body diagram of forces is constructed to determine the forces acting on different parts of the machine.

Step 3 - Selection of Material

- Four basic factors, which are considered in selecting the material, are availability, cost, Mechanical properties & manufacturing considerations.

Step 4 - Failure Criterion

- It is necessary to know the type of failure by which the component will fail when put into the service.
- The three basic types of failure are as follows:
 - i) Failure by elastic deflection
 - ii) Failure by general yielding &
 - iii) Failure by fracture.

Step 5 - Determination of Dimensions

- Determination of geometric dimensions is an important step in design of machine elements.
- Various criteria such as yield strength, ultimate tensile strength, torsional or lateral deflection, & permissible bearing pressure are used to find out these dimensions.

Step 6 - Design Modifications

- The geometric dimensions of the machine element are modified from assembly & manufacturing considerations.

Step 7 - Working Drawing

- The working drawing must be clear, concise & complete. It must have enough views & cross-sections to show all details.

Q.2 State & explain various steps involved in Machine design.

→ The following steps are involved in the process of machine design,

Market Survey



Design Define Specifications of Product



Study Alternative Mechanism for Product
& Select Proper mechanism



Prepare general layout of Configuration
& select Joining methods between
Individual Components of Product



Design Individual Components



Prepare Assembly & Detail Drawings
& Modify Drawing after Testing
Proto type Model

Q.3 Note on Aesthetic Consideration in design.

- - When there are a number of products in the market, having the same qualities of efficiency, durability of cost, the customer is attracted towards the most appealing product.
- The external appearance is an important feature, which not only gives grace & lustre to the product but also dominates the sale of in market.
- This is particularly true for customer durables like automobiles, household appliances & audio visual equipment.
- The growing realization of the need of aesthetic considerations in product design has given rise to a separate discipline, known as industrial design.
- The job of industrial designer is to create new forms & shapes, which are aesthetically pleasing.
- There are five basic forms- step, stream, taper, shear & sculpture. The step form is similar to the shape of a 'skyscraper' or multi-storey building. This involves shapes with a vertical accent rather than a horizontal.
- The stream or streamline form is seen in automobiles & aeroplane structures.
- The taper form consists of tapered blocks interlocked with tapered plinths or cylinders.
- The shear form has a square outlook, which is suitable for free standing engineering products.
- The sculpture form consists of ellipsoids, paraboloids & hyperboloids.
- The sculpture & stream forms are suitable for mobile products like vehicles, while step & shear forms for stationary products.

- Selection of proper colour is an important consideration in product aesthetics.
- The choice of colour should be compatible with the conventional ideas of the operator.
- Many colours are associated with different moods & conditions.
- Morgan has suggested the meaning of colours that are given in below Table

Colour	Meaning
Red	Danger-Hazard-Hot
Orange	Possible danger
Yellow	Caution
Green	Safety
Blue	Caution- cold
Grey	Dull

- The external appearance of the product does not depend upon only two factors - form & colour.
- It is a cumulative effect of number of factors such as rigidity & resilience, tolerances & surface finish, motion of individual components, materials, manufacturing methods & noise.
- The industrial designer should select a form, which is in harmony with the functional requirement of the product.
- The economics & availability of surface treating processes like anodizing, plating, blackening & painting should be taken into account before finalizing the external appearance of the product.

Q.5 How material is selected while designing machine element (Factors influencing selection of material)

→ * Selection of Material

- Four basic factors, which are considered in selecting the material, are availability, cost, mechanical properties, & manufacturing considerations.
- For example, Flywheel, housing of gear box, or engine block have complex shapes.
- These components are made of cast iron because casting process produces complicated shapes without involving machining operations.
- Transmission shafts are made of plain carbon steels, because they are available in the form of rods, besides their higher strength.
- The automobile body & hood are made of low carbon steels because their cold formability is essential to press the parts.
- Free cutting steels have excellent machinability due to addition of Sulphur.
- They are ideally suitable for bolts & studs because of the ease with which the thread profiles can be machined.
- Crank-shaft & connecting rod are subjected to fluctuating forces & nickel-chromium steel is used for these components due to its higher fatigue strength.

Note On Preferred Series.

- In engineering design, many times the designer has to specify the size of the product.
- The 'size' of the product is a general term, which includes different parameters like power transmitting capacity, load carrying capacity, speed & dimensions of the component such as height, length & width, & volume or weight of the product.
- These parameters are expressed numerically, e.g. 5kW, 10kN or 1000 rpm.
- Often, the product is manufactured in different sizes or models, for instance, a company may be manufacturing seven different models of electric motors ranging from 0.5kW to 50 kW to cater to the need of different customers.
- Preferred numbers are used to specify the 'sizes' of the product in these cases.
- French balloonist & engineer Charles Renard first introduced preferred numbers in the 19th century.
- The system is based on the use of geometric progression to develop a set of numbers.
- There are five basic series, denoted as R5, R10, R20, R40 & R80 series, which increase in steps of 58%, 26%, 12%, 6% & 3% resp.
- Each series has its own series factor.

- The series factors are given in below table

R5 Series	$\sqrt[5]{10} = 1.58$
R10 Series	$\sqrt[10]{10} = 1.26$
R20 Series	$\sqrt[20]{10} = 1.12$
R40 Series	$\sqrt[40]{10} = 1.06$
R80 Series	$\sqrt[80]{10} = 1.03$

- The series is established by taking the first number & multiply multiplying it by a series factor, to get the second number.
- The second number is again multiplied by a series factor to get the third number.
- This procedure is continued until the complete series is built up.
- The resultant numbers are rounded & s.
- As an example, consider a manufacture of lifting tackles who wants to introduce nine different models of capacities ranging from about 15 to 100 kN.
- Referring to R10 series, the capacities of different models of lifting tackle will be 16, 20, 25, 31.5, 40, 50, 63, 80 & 100 kN.